CO2-AT 3

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Computer vision

1) Image Enhancement in the Spatial Domain.

Image enhancement is the process of improving the visual quality of an image or making important details easier to identify. In the spatial domain, enhancement is performed directly on the pixels of an image.

$$g(x,y) = T[f(x,y)].$$

(i) Gray Level Transformations:

* Modify individual pixel intensity values.
* Negative transformation: reverses brightness
* log transformation: enhances dark regions
* Power-law transformation: adjust brightness.
* Contrast Stretching: expands a narrow intensity range.

(ii) Histogram Processing:

* Histogram represents the frequency of different intensity levels.

* Histogram equalization redistributes values to impr[ove] contrast.
* It is useful for low-contrast images

(ii) Spatial Filtering:

* Uses a mask/kernel on neighboring Pixels.
* Smoothing filters such as mean and Gaussian reduce noise.
* Sharpening filters such as Laplacian and Sobel enhance edges.

These techniques improve captured Images and prepare them for further processing such as segmentation and object detection.

2) Image Smoothing and Sharpening:

Smoothing reduces noise and unwanted details while sharpening enhances edges and fine details.

Smoothing:

A smoothing filter replaces a pixel using its neighboring Pixels.

Gaussian filter

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

It gives more importance to nearby pixels and reduces noise.

Used when: Image contain noise or before edge detection.

Sharpening:

Sharpening emphasizes sudden changes in Intensity.

Laplacian filter:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Used when: Images are blurred or object boundaries need enhancement.

Smoothing is mainly for noise removal, whereas Sharpening is mainly for edge and detail enhancement.

3) Frequency Domain Enhancement:

Frequency-domain enhancement processes an image using its frequency components.

Process

$f(x,y) \rightarrow$ Fourier Transform $\rightarrow F(u,v)$.

A filter is applied:

$$G(u,v) = H(u,v)\,F(u,v)$$

Then Inverse Fourier transform produces the enhanced image.

Low-Pass Filter (LPF):

Allows low frequency and removes high frequencies.

Effects:

- Reduces noise.
- Smooths the image.
- Reduces fine details.

Applications:

Noise removal and image smoothing.

High-Pass Filter (HPF)

Allows high frequencies and suppresses low frequencies.

Effects

* Enhances edges
* Improves sharpness
* Highlights fine details

Applications: Edge enhancement and feature extraction.

LPF	HPF
Smoothing	Sharpening
Reduces noise	Enhances edge
Removes high frequencies	Enhances high frequencies

LPF is suitable for noise reduction, while HPF is suitable for edge and detail enhancement.

4) Image segmentation.

Image segmentation divides an Image into meaningful regions or objects

Main techniques:

1. Point Detection:

* Detects isolated pixels that differ from their surroundings.
* Advantage: Simple and fast
* Limitation: Sensitive to noise
* Application: Small defect detection

2. Line Detection:

* Detects lines of particular orientations.
* Advantage: Detects structural lines
* Limitation: Direction dependent
* Application: Road and document analysis.

3. Edge Detection:

* Detects sudden changes in Intensity.
* Uses Sobel, Prewitt, Roberts and Canny.
* Detects boundaries and defect detection.

* Thresholding
* Separates foreground and background using threshold T
* Advantage: Simple and fast
* Limitation: Sensitive to lighting.
* Application: Document and object extraction.

5. Region-Based Segmentation:
* Groups neighboring pixels having similar properties.
* Advantage: Produces connected regions
* Limitation: Can require more computation
* Application: Medical and object segmentation

5) Edge Detection, Edge Linking and Boundary Detection.

Edge Detection:

Edge detection identifies areas where there is a sudden change in intensity.

Common operators:

* Sobel
* Prewitt
* Roberts
* Canny.

For Sobel:

$$G = \sqrt{G_x^2 + G_y^2}$$

Edge linking:

Edge detection may produce broken edges. Edge linking connects related edge pixels to form continuous edges.

Broken: — — —

After linking: ———

It improves the continuity of object boundaries.

Boundary Detection:

Boundary detection identifies the complete contour of a object.

Importance:

* object recognition.
* Shape analysis
* Crack and Scratch detection
* Industrial defect detection.
* Medical Image analysis.

Edge detection find edges, edge linking connect broken edges, and boundary detection identifies complete object contour and boundary detection identifies